Recurrence Relations

Sequences that appear in mathematics and science are frequently defined by recurrences, rather than by formulas. A variety of techniques have been developed to obtain explicit formulas for the values in some cases. We will look for solutions of the recurrences of the form:

$s_n = as_{n-1} + bs_{n-2}$ where a and b are constants.

Ex:

Consider the recurrence relation $s_n = 3s_{n-1}$ with $s_0 = 5$. Here $a = 5$ and $b = 0$.

Then, $s_1 = 5 \cdot 3^1$ for $n \in \mathbb{N}$.

Ex:

Consider the recurrence relation $s_n = 3s_{n-2}$ with $s_0 = 5$ and $s_1 = 2$. Here $a = 0$ and $b = 5$.

Then, $s_{2n+1} = 2 \cdot 3^n$ $\quad$ $s_{2n} = 5 \cdot 3^n$

From these special cases we have examined, it seems reasonable to hope that some solutions have the form $s_n = c.r^n$ for a nonzero constant c. This hope, if true, would force us to the fact that

$r^n = a.r^{n-1} + b.r^{n-2}$ ➔ $r^2 = a.r + b$ ➔ $r^2 - a.r - b = 0$

If it turns out that $s_n = c.r^n$ for all n, then r must be a solution of the quadratic equation $x^2 - a.x - b = 0$, which is called the characteristic equation of the recurrence relation.

Theorem 1:

Consider a recurrence relation of the form

$s_n = as_{n-1} + bs_{n-2}$ with characteristic equation $x^2 - a.x - b = 0$ where a and b are nonzero constants.

- If the characteristic equation has 2 different solutions r_1 and r_2, then $s_n = c_1.r_1^n + c_2.r_2^n$ for certain constants c_1 and c_2.
- If the characteristic equation has only one solution r, then $s_n = c_1.r^n + c_2.n.r^n$ for certain constants c_1 and c_2.

Ex:

Consider $s_n = s_{n-1} + 2.s_{n-2}$ where $s_0 = 3$ and $s_1 = 3$.

Characteristic Equation: $x^2 - x - 2 = 0 \Rightarrow (x+1).(x-2) = 0$

$r_1 = -1$, $r_2 = 2$.

$s_n = c_1 \cdot 2^n + c_2 \cdot (-1)^n$

$s_0 = 3 = c_1 \cdot 2^0 + c_2 \cdot (-1)^0 = c_1 + c_2$

$s_1 = 3 = c_1 \cdot 2^1 + c_2 \cdot (-1)^1 = 2c_1 - c_2$

$c_1 + c_2 = 3$, $2c_1 - c_2 = 3 \Rightarrow c_1 = 2$, $c_2 = 1$

$s_n = 2 \cdot 2^n + 1 \cdot (-1)^n$ for $n \in \mathbb{N}$.

Fibonacci Series:

Ex: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

s_0 s_1 s_2 s_3 s_4 s_5 s_6 s_7 s_8 s_9 s_{10} s_{11}, ...

Consider Fibonacci series: $s_n = s_{n-1} + s_{n-2}$ for $n \geq 2$, $s_0 = 0$ and $s_1 = 1$.

$ax^2 + bx + c = 0$

$x^2 - x - 1 = 0$

$r_1 = \frac{1+\sqrt{1+4}}{2} = \frac{1+\sqrt{5}}{2}$, $r_2 = \frac{1-\sqrt{5}}{2}$

$x_{1,2} = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$

$s_n = c_1 \cdot r_1^n + c_2 \cdot r_2^n = c_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$ for $n \in \mathbb{N}$

$s_0 = 0 = c_1 \cdot 1 + c_2 \cdot 1 \Rightarrow c_1 + c_2 = 0$

$s_1 = 1 = c_1 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^1 + c_2 \cdot \left(\frac{1-\sqrt{5}}{2}\right)^1$

Solve for c_1 and c_2

$c_1 = \frac{1}{\sqrt{5}}$ and $c_2 = \frac{-1}{\sqrt{5}}$

$s_n = c_1 \cdot r_1^n + c_2 \cdot r_2^n = \frac{1}{\sqrt{5}} r_1^n - \frac{1}{\sqrt{5}} r_2^n = \frac{1}{\sqrt{5}} \left[r_1^n - r_2^n\right]$ for $n \geq 0$

So, $Fib(n) = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$

Ex:

Consider $s_n = 6.s_{n-1} - 9.s_{n-2}$ for $n \geq 2$, $s_0 = 1$ and $s_1 = -3$.

C.E: $x^2 - 6x + 9 \Rightarrow r_{1,2} = 3 = r$

$s_n = c_1 \cdot 3^n + c_2 \cdot n \cdot 3^n$ for $n \in \mathbb{N}$

For $n=0$: $s_0 = c_1 \cdot 3^0 + c_2 \cdot 0 \cdot 3^0 = c_1 = 1$

For $n=1$: $s_1 = c_1 \cdot 3^1 + c_2 \cdot 1 \cdot 3^1 = 3c_1 + 3c_2 = -3 \Rightarrow c_2 = -2$

Therefore $s_n = 3^n - 2 \cdot n \cdot 3^n$

Ex:

Consider $s_n = s_{n-1} + 2.s_{n-2}$ for $n \geq 2$, $s_0 = 3$ and $s_1 = 6$.

$x^2 - x - 2 = 0 \Rightarrow r_1 = 2, \quad r_2 = -1$

$s_n = c_1 \cdot 2^n + c_2 \cdot (-1)^n$

$s_0 = 3 = c_1 \cdot 2^0 + c_2 (-1)^0 = c_1 + c_2 = 3$

$s_1 = 6 = c_1 \cdot 2^1 + c_2 (-1)^1 = 2c_1 - c_2 = 6$

$\Rightarrow c_1 = 3, \quad c_2 = 0$

Then, $s_n = 3 \cdot 2^n$

I switch to using a_n instead of S_n from now on.

Nonhomogeneous Recurrence Relations

Let

(1) $a_n + xa_{n-1} = f(n)$ for $n \geq 1$

(2) $a_n + xa_{n-1} + ya_{n-2} = f(n)$ for $n \geq 2$

$S_n + x \cdot S_{n-1} = \not{0}$ $\quad S_n = -x \cdot S_{n-1}$

where x and y are constants, $x \neq 0$ in (1), $y \neq 0$, an $f(n)$ is not identically 0.

Although there is no general method for solving all nonhomogeneous relations, for certain functions $f(n)$ we shall find a successful technique.

Consider a special case when x=-1 in (1). Then, for $a_n - a_{n-1} = f(n)$, we have

$a_1 = a_0 + f(1)$,

$a_2 = a_1 + f(2) = a_0 + f(1) + f(2)$,

$a_3 = a_2 + f(3) = a_0 + f(1) + f(2) + f(3)$,

$\vdots$

$a_n = a_0 + f(1) + \cdots + f(n) = a_0 + \sum_{i=1}^{n} f(i)$,

We can solve this type of relation in terms of n, if we can find a suitable summation formula for $\sum_{i=1}^{n} f(i)$.

Ex: Solve $a_n - a_{n-1} = 3n^2$ $\quad a_0 = 7$

Observe that $f(n) = 3n^2$ $\quad f(n) = 3 \cdot n^2$

Then $a_n = a_0 + \sum_{i=1}^{n} f(i) = 7 + 3 \cdot \sum_{i=1}^{n} i^2 = 7 + \frac{3 \cdot n \cdot (n+1) \cdot (2n+1)}{6}$

$= 7 + \frac{n \cdot (n+1) \cdot (2n+1)}{2}$

$\frac{n \cdot (n+1) \cdot (2n+1)}{6} =$

For either (1) or (2), we let $a_n^{(h)}$ denote the general solution of the associated homogeneous relation, and we let $a_n^{(p)}$ be a solution of the given nonhomogeneous relation. The term $a_n^{(p)}$ is called a particular solution. Then $a_n = a_n^{(h)} + a_n^{(p)}$ is the general solution of the given relation. To determine $a_n^{(p)}$ we use the form of $f(n)$ to suggest a form $a_n^{(p)}$.

Ex: Solve $a_n - 3a_{n-1} = 5 \cdot (7^n),\ n \geq 1, a_0 = 2$

Corresponding homogeneous relation is: $a_n - 3a_{n-1} = 0$

Solution to this associated hom. relation $\rightarrow$ $a_n^{(h)} = C \cdot 3^n$

Since $f(n) = 5(7^n)$ we seek a particular solution $a_n^{(p)}$ of the form $A \cdot (7^n)$.

Then, $A \cdot (7^n) - 3 \cdot A \cdot (7^{n-1}) = 5 \cdot (7^n)$

$A \cdot 7 - 3A = 5 \cdot 7 \Rightarrow A = \frac{35}{4}$

Then, $a_n^{(p)} = \frac{35}{4} \cdot 7^n = \frac{5}{4} \cdot 7^{n+1}$

Hence, the general solution is $a_n = C \cdot 3^n + \frac{5}{4} \cdot 7^{n+1}$

With $a_0 = 2$: $2 = C + \frac{35}{4} \Rightarrow C = \frac{-27}{4}$

Therefore $a_n = \frac{5}{4}(7^{n+1}) - \frac{1}{4}(3^{n+3})$

Ex: Solve $a_n - 3a_{n-1} = 5 \cdot (3^n),\ n \geq 1, a_0 = 2$

$a_n^{(h)} = C \cdot 3^n$ But here $a_n^{(h)}$ and $f(n)$ are not linearly independent.

As a result, we consider $a_n^{(p)}$ of the form $B \cdot n \cdot 3^n$ ~~$B \cdot 3^n$~~

Then, $Bn\,3^n - 3B(n-1) \cdot 3^{n-1} = 5 \cdot 3^n$

$Bn - B(n-1) = 5 \Rightarrow B = 5$

Hence, $a_n = a_n^{(h)} + a_n^{(p)} = C \cdot 3^n + 5 \cdot n \cdot 3^n = (C + 5n) \cdot 3^n$ ← general solution

use a_0:

$a_0 = 2 \Rightarrow a_n = (2 + 5n) \cdot 3^n$

From these examples, we can generalize as follows:

If $a_n + c_1 a_{n-1} + c_2 a_{n-2} = k \cdot (r^n)$, where c_1, c_2 and k are constants.

Then, we find that:

- $a_n^{(p)} = A \cdot (r^n)$ for A a constant, if r^n is not a solution of the associated hom. relation,
- $a_n^{(p)} = B \cdot n \cdot (r^n)$ for B a constant, if $a_n^{(h)} = d_1 r^n + d_2 r_1^n$ where $r \neq r_1$,
- $a_n^{(p)} = C \cdot n^2 \cdot (r^n)$ for C a constant, if $a_n^{(h)} = (c_1 + c_2 n) r^n$.

Ex: Solve $a_{n+1} = 2a_n + 1$, $n \geq 0$, $a_0 = 0$

Corresponding hom. relation: $a_{n+1} - 2a_n = 0$

Then, $a_n^{(h)} = C \cdot 2^n$. Observe that $f(n) = 1 = 1^n$ is not a solution of associated hom. relation.

Hence, we can set $a_n^{(p)} = A \cdot 1^n = A$.

Find that $A = 2A + 1 \Rightarrow A = -1$

Then, the general solution is $a_n = a_n^{(h)} + a_n^{(p)} = C \cdot 2^n - 1$

From $a_0 = 0 \Rightarrow C - 1 = 0 \Rightarrow C = 1$.

Therefore, $a_n = 2^n - 1$

Ex: Solve $a_{n+1} = 2a_n + 2^n,\ n \geq 1,\ a_1 = 1$

$a_{n+1} - 2a_n = 0$ ← hom. relation

$a_n^{(h)} = C \cdot 2^n$

observe that $f(n) = 2^n$ is a solution of $a_{n+1} - 2a_n = 0$.

Here, we set $a_n^{(p)} = A \cdot n \cdot 2^n$ and find that

$A \cdot (n+1) \cdot 2^{n+1} = 2 \cdot A \cdot n \cdot 2^n + 2^n$

$2 \cdot A \cdot n \cdot 2^n + 2 \cdot A \cdot 2^n = 2A \cdot n \cdot 2^n + 2^n$

$2An + 2A = 2An + 1 \Rightarrow 2A = 1 \Rightarrow A = 1/2$.

Then, $a_n^{(p)} = \frac{1}{2} \cdot n \cdot 2^n = n \cdot 2^{n-1}$.

Then, $a_n = C \cdot 2^n + n \cdot 2^{n-1}$ || from $a_1 = 1 \Rightarrow 1 = C \cdot 2 + 1 \cdot 1$

$1 = 2C + 1 \Rightarrow C = 0$

Here, $a_n = n \cdot 2^{n-1}$

Ex: Solve $a_{n+2} - 4a_{n+1} + 3a_n = -200,\ n \geq 0,\ a_0 = 3000,\ a_1 = 3300.$

$a_n^{(h)} = d_1 3^n + d_2 1^n = d_1 3^n + d_2$

$a_{n+2} - 4a_{n+1} + 3a_n = 0$

$(x-3)(x-1) = 0$

$r_1 = 3$

$r_2 = 1$

Since $f(n) = -200 = -200 \cdot (1^n)$ is a solution of the hom. relation we set $a_n^{(p)} = A \cdot n$.

Then $A(n+2) - 4 \cdot A(n+1) + 3 \cdot A \cdot n = -200$

$An + 2A - 4An - 4A + 3An = -200 \Rightarrow -2A = -200$

$A = 100$

Here, $a_n = d_1 \cdot 3^n + d_2 + 100n$ ← general solution

With $a_0 = 3000$ and $a_1 = 3300$ we obtain

$a_n = 100 \cdot 3^n + 2900 + 100n$

We now summarize and extend the solution techniques already discussed.

Given a linear nonhomogeneous recurrence relation (with constant coefficients) of the form $d_1a_n + d_2a_{n-1} + \cdots + d_{k+1}a_{n-k} = f(n)$, $d_1 \neq 0$ and $d_{k+1} \neq 0$. Let $a_n^{(h)}$ denote the homogeneous part of the solution a_n.

1. If $f(n)$ is a constant multiple of one of the forms in the first column of the table and is not a solution of the associated homogeneous relation, then $a_n^{(p)}$ has the form shown in the second column. (Here $A, A_1, A_2 \ldots A_{t-1}, A_t$ are constants determined by substituting $a_n^{(p)}$ into the given relation; t and r are also constants.)
2. When $f(n)$ comprises a sum of the constant multiples of terms such as those in the first column of the table, and none of these terms is a solution of the associated homogeneous relation, then $a_n^{(p)}$ is made up of the sum of the corresponding terms in the column headed by $a_n^{(p)}$. For example if $f(n) = n^2 + 3(2^n)$ and no summand of the $f(n)$ is a solution of the associated homogeneous relation, then $a_n^{(p)} = (A_2n^2 + A_1n + A_0) + A(2^n)$
3. Consider the case where if a summand $f_1(n)$ of $f(n)$ is a solution of the associated homogenous relation. This happens, for example, when $f(n)$ contains summands such as cr^n or $(c_1 + c_2n)r^n$ and r is a characteristic root. If $f_1(n)$ causes the problem, we multiply the trial solution $a_{n_1}^{(p)}$ corresponding to $f_1(n)$ by the smallest power of n, say n^s, for which no summand of $n^s f_1(n)$ is a solution of the associated homogeneous relation. Then $n^s a_{n_1}^{(p)}$ is the corresponding part of $a_n^{(p)}$

$f(n)$	$a_n^{(p)}$
$c, a\ constant$	$A, a\ constant$
n	$A_1n + A_0$
n^2	$(A_2n^2 + A_1n + A_0)$
$n^t, t \in \mathbb{Z}^+$	$A_tn^t + A_{t-1}n^{t-1} + \ldots + A_1n + A_0$
$r^n, r \in \mathbb{R}$	Ar^n
n^tr^n	$r^n(A_tn^t + A_{t-1}n^{t-1} + \ldots + A_1n + A_0)$

Ex: Solve $a_{n+1} = a_n + n,\ n \geq 2, a_2 = 1$

$a_n^{(h)} = c \cdot (1^n) = c$. Since the hom. particular solution for $f(n) = n$ is $A_1 n + A_0$ and A_0 in $(A_1 n + A_0)$ is a solution of the associated hom. relation (i.e. they are linearly dependent) we multiply $A_1 n + A_0$ by the smallest power of n for which we no longer have any constant summand. We accomplish this by multiplying $(A_1 n + A_0)$ by n^1.

So, $a_n^{(p)} = A_1 \cdot n^2 + A_0 \cdot n$. Then, $A_1(n+1)^2 + A_0(n+1) = A_1 n^2 + A_0 n + n$

Then $A_1 n^2 + A_1 \cdot 2n + A_1 + A_0 n + A_0 = A_1 n^2 + A_0 n + n$

$\Rightarrow$ $2A_1 = 1 \Rightarrow A_1 = \frac{1}{2}$

$A_1 + A_0 = 0 \Rightarrow A_0 = -\frac{1}{2}$

So, $a_n^{(p)} = \frac{1}{2}n^2 - \frac{1}{2}n$ | Hence, $a_n = \frac{1}{2}n^2 - \frac{1}{2}n + c$

using $a_2 = 1 \Rightarrow 1 = \frac{1}{2}\cdot 4 - \frac{1}{2}\cdot 2 + c \Rightarrow c = 0$. Therefore, $a_n = \frac{1}{2}n^2 - \frac{1}{2}n$

Ex: Consider $a_{n+2} - 10a_{n+1} + 21a_n = f(n)$.

The homogeneous solution is $c_1(3^n) + c_2(7^n)$ for constants c_1 and c_2.

What is $a_n^{(p)}$ if *f(n)* is

a) 5 $\longrightarrow$ $a_n^{(p)} = A_0$

b) $3n^2 - 2$ $\longrightarrow$ $a_n^{(p)} = A_2 \cdot n^2 + A_1 \cdot n + A_0$

c) $7(11^n)$ $\longrightarrow$ $a_n^{(p)} = A_4 \cdot (11^n)$

d) $31\,(r^n), r \neq 3,7$ $\longrightarrow$ $a_n^{(p)} = A_5 \cdot (r^n)$

e) $6(3^n)$ $\longrightarrow$ $a_n^{(p)} = A_6 \cdot n \cdot 3^n$

f) $2(3^n) - 8(9^n)$ $\longrightarrow$ $a_n^{(p)} = A_7 n \cdot 3^n - A_8 \cdot 9^n$

g) $4(3^n) + 3(7^n)$ $\longrightarrow$ $a_n^{(p)} = A_9 \cdot n \cdot 3^n + A_{10} \cdot n \cdot 7^n$

Ex: Consider $a_n + 4a_{n-1} + 4a_{n-2} = f(n)$, $n \geq 2$

The homogeneous solution is $c_1(-2)^n + c_2 n(-2)^n$ Then,

If $f(n) = 5.(-2)^n$, then $a_n^{(p)} = A \cdot n^2 \cdot (-2)^n$

If $f(n) = 7.n.(-2)^n$, then $a_n^{(p)} = n^2 \cdot (-2)^n \cdot (B_1 \cdot n + B_0)$

If $f(n) = -11.n^2.(-2)^n$, then $a_n^{(p)} = n^2 \cdot (-2)^n \cdot (C_2 \cdot n^2 + C_1 \cdot n + C_0)$

Here $A, B_0, B_1, C_0, C_1, C_2$ are determined by substituting $a_n^{(p)}$ into given nonhomogeneous relation.